

Experimental and DFT-Assisted Raman Characterisation of N-Acetylmuramic Acid for Bacterial Biomarker Analysis

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Abstract—N-acetylmuramic acid (NAM) is an important structural component of bacterial peptidoglycan and plays a significant role in studies related to bacterial identification and biomarker-based diagnostics. In the present work, the vibrational characteristics of pure N-acetylmuramic acid powder were investigated using experimental Raman spectroscopy combined with density functional theory (DFT)-based computational analysis. Raman spectra were acquired in the spectral range of 500–2000 cm^{-1} using a 785 nm excitation source under optimised acquisition conditions. To improve spectral quality, preprocessing steps including dark subtraction, baseline correction, and Savitzky–Golay smoothing were performed prior to analysis.

The experimental Raman spectra were compared with theoretically simulated vibrational frequencies obtained using DFT calculations carried out in Gaussian software employing the B3LYP functional and 6-31G(d) basis set. Detailed vibrational assignments were performed for the major Raman bands corresponding to carbohydrate ring vibrations, C–O stretching, CH bending, and acetyl-associated vibrational modes. A good agreement between the experimental and simulated spectra was observed, supporting the reliability of the theoretical model and the proposed peak assignments.

The study establishes reference Raman fingerprints of isolated N-acetylmuramic acid and provides experimentally validated vibrational information relevant to bacterial cell-wall biomarker analysis. The obtained results may contribute toward future Raman- and SERS-based bacterial sensing studies, biomolecular spectral databases, and computational vibrational spectroscopy investigations involving peptidoglycan-related molecules.

Index Terms—Raman spectroscopy, Density functional theory, N-acetylmuramic acid, Bacterial biomarkers, Peptidoglycan, Vibrational spectroscopy, SERS

I. INTRODUCTION

Raman spectroscopy has emerged as a powerful vibrational spectroscopic technique for the characterization of biological molecules due to its non-destructive, label-free, and highly sensitive molecular fingerprinting capability. In recent years, Raman-based approaches have gained considerable attention in biomedical diagnostics, biomolecular identification, and bacterial detection studies because they provide detailed information about molecular structure and chemical composition. The technique has been widely applied in the analysis of proteins, carbohydrates, nucleic acids, and complex biological systems, including microbial cell-wall components and bacterial metabolites. In addition, the advancement of computational vibrational analysis has further improved the interpretation

of Raman spectra through theoretical prediction and peak assignment studies.

Among bacterial cell-wall constituents, N-acetylmuramic acid (NAM) is an important amino sugar molecule present in the peptidoglycan layer of both Gram-positive and Gram-negative bacteria. NAM, together with N-acetylglucosamine (NAG), forms the glycan backbone responsible for maintaining bacterial cell-wall rigidity and structural integrity. Due to its direct association with bacterial peptidoglycan, NAM has attracted interest in bacterial biomarker research, microbial identification, and biosensing applications. The molecular vibrational characteristics of NAM are therefore important for understanding bacterial spectral signatures and for developing future Raman- and SERS-based diagnostic approaches targeting bacterial infections and sepsis-related biomarkers.

Several Raman spectroscopic studies have previously focused on whole bacterial systems, amino sugars, and peptidoglycan-related biomolecules. Experimental Raman investigations of glucosamine and N-acetylglucosamine have provided valuable insights into carbohydrate vibrational modes and amino sugar spectral features. Similarly, Raman and SERS studies on bacterial samples have reported characteristic spectral bands associated with bacterial cell-wall components and metabolic signatures. Computational approaches based on density functional theory (DFT) have also been extensively used for the prediction of vibrational frequency and spectral assignment of biologically relevant molecules. However, despite the biological importance of NAM, comprehensive experimental Raman characterization and detailed DFT-assisted vibrational assignment studies of isolated N-acetylmuramic acid remain limited in the existing literature. Most reported studies investigate whole bacterial systems or focus on structurally related molecules rather than isolated NAM powder.

In this work, an experimental and theoretical vibrational spectroscopic investigation of pure N-acetylmuramic acid powder was carried out using Raman spectroscopy combined with density functional theory calculations. Experimental Raman spectra were obtained under optimized acquisition conditions and compared with theoretically simulated vibrational frequencies generated using Gaussian software. Detailed vibrational assignments were performed for the major Raman bands that correspond to characteristic molecular vibrations of NAM.

The purpose of the present study was to establish experimentally validated Raman fingerprints of isolated N-acetylmuramic acid and to provide reference vibrational information relevant to bacterial biomarker analysis, computational spectroscopy, and future Raman/SERS-based biosensing applications.

II. MATERIALS AND METHODS

A. Materials

Pure N-acetylmuramic acid (NAM) powder was used as the primary analyte for the present study. The powder sample was used without any additional chemical modification or purification. For Raman spectroscopic measurements, a small quantity of the NAM powder was directly placed onto a clean glass substrate/sample holder for spectral acquisition. Care was taken to avoid contamination and moisture exposure during sample preparation and handling.

The molecular structure of N-acetylmuramic acid consists of carbohydrate and acetyl-associated functional groups relevant to bacterial peptidoglycan systems. Due to its importance as a structural component of bacterial cell walls, the molecule was selected for vibrational spectroscopic characterization and theoretical spectral analysis.

All experimental measurements were carried out under ambient laboratory conditions. The obtained Raman spectra were further processed and analyzed for comparison with theoretically simulated vibrational spectra generated using density functional theory calculations performed in Gaussian software.

B. Raman Measurements

Raman spectroscopic measurements were carried out using a Raman microscopy system equipped with a 785 nm excitation laser source. A small quantity of pure powder was placed on a clean sample holder and analyzed under ambient laboratory conditions. Spectra were recorded in the spectral range of 500–2000 cm^{-1} to capture the characteristic vibrational modes associated with the molecular structure of N-acetylmuramic acid.

To optimize spectral quality and minimize noise, measurements were performed under different laser power conditions while maintaining a constant acquisition setting. Raman spectra were acquired with an integration time of 10 s and 10 accumulations for each measurement. Multiple spectra were collected from different regions of the sample to ensure spectral consistency and reproducibility. Care was taken to avoid thermal damage and fluorescence interference during laser exposure.

The acquired spectra were stored for further preprocessing and comparative analysis with the theoretically simulated Raman spectra obtained from density functional theory calculations. The experimental Raman measurements were primarily focused on identifying characteristic vibrational bands associated with carbohydrate ring vibrations, C–O stretching, CH bending, and acetyl-related molecular vibrations of N-acetylmuramic acid.

C. Spectral Preprocessing

The acquired Raman spectra of N-acetylmuramic acid were subjected to preprocessing prior to spectral interpretation and comparison with the theoretical simulations. Initial preprocessing involved dark signal subtraction to reduce background noise contributions originating from the instrument and detector system. The spectra were then visually inspected to remove any abnormal or corrupted spectral acquisitions.

To correct spectral baseline distortions and fluorescence background effects, asymmetric least squares (ALS) baseline correction was applied to the raw Raman spectra. Following baseline correction, Savitzky–Golay smoothing was performed to improve signal quality while preserving the original spectral features and peak positions. The preprocessing parameters were selected carefully to avoid artificial distortion of Raman peaks.

The processed spectra were subsequently normalized for comparative analysis between different acquisition conditions and for comparison with the density functional theory (DFT)-simulated Raman spectra generated using Gaussian software. The final preprocessed spectra were used for peak identification, vibrational assignment, and experimental–theoretical spectral correlation analysis.

D. DFT Calculations

Theoretical vibrational analysis of N-acetylmuramic acid was carried out using density functional theory (DFT) calculations implemented in Gaussian software. The molecular geometry of N-acetylmuramic acid was initially optimized to obtain the energetically stable structure prior to vibrational frequency calculations. All calculations were performed using the B3LYP exchange–correlation functional in combination with the 6-31G(d) basis set, which is widely used for vibrational spectroscopic investigations of biomolecular systems.

Following geometry optimization, vibrational frequency calculations were performed to obtain the theoretical Raman-active vibrational modes and corresponding Raman activities of the molecule. The absence of imaginary frequencies confirmed that the optimized molecular structure corresponded to a stable minimum on the potential energy surface. The calculated vibrational frequencies were further analyzed and compared with the experimentally obtained Raman spectra.

To improve agreement between theoretical and experimental frequencies, appropriate frequency scaling was considered during spectral comparison. The simulated Raman spectra generated from the DFT calculations were used for vibrational band assignment and identification of characteristic molecular vibrations associated with carbohydrate ring modes, C–O stretching vibrations, CH bending vibrations, and acetyl-related functional groups present in N-acetylmuramic acid.

The theoretical calculations provided molecular-level insight into the vibrational behavior of the analyte and supported the interpretation of the experimentally observed Raman spectral features.

III. RESULTS AND DISCUSSION

A. Experimental Raman Spectrum

The experimental Raman spectrum of pure N-acetylmuramic acid powder was successfully acquired in the spectral range of 500–2000 cm^{-1} using a 785 nm excitation laser source. The obtained spectrum exhibited several characteristic vibrational bands corresponding to the molecular structure and functional groups present in N-acetylmuramic acid. The Raman signals obtained from the powder sample showed good spectral consistency and reproducibility across multiple measurements performed under optimized acquisition conditions.

The recorded Raman spectrum contained prominent peaks associated with carbohydrate ring vibrations, C–O stretching modes, CH bending vibrations, and acetyl-related molecular vibrations. Strong Raman activity was observed in the fingerprint region, particularly between 800–1500 cm^{-1} , which is typically associated with vibrational modes originating from amino sugar and carbohydrate-based molecular structures. The spectral region corresponding to 1000–1200 cm^{-1} exhibited characteristic carbohydrate-related vibrational features, while bands observed near the higher wavenumber regions were associated with CH deformation and acetyl functional group vibrations.

The acquired spectra were initially affected by minor baseline distortions and background fluorescence contributions, which were minimized through preprocessing procedures including baseline correction and smoothing. After preprocessing, the Raman peaks became more distinguishable and suitable for further vibrational analysis and comparison with the theoretically simulated spectra obtained from density functional theory calculations.

Variations in laser power conditions were also examined during spectral acquisition to optimize signal quality while avoiding thermal degradation of the sample. Lower laser powers produced weaker Raman intensity, whereas higher laser powers improved signal strength but required careful optimization to prevent excessive background interference and possible thermal effects. Optimized acquisition parameters provided clear and stable Raman spectral signatures suitable for detailed vibrational assignment studies.

The experimentally obtained Raman spectrum serves as a reference vibrational fingerprint of isolated N-acetylmuramic acid and provides important spectral information relevant to bacterial peptidoglycan studies and future Raman- and SERS-based bacterial biomarker detection applications.

B. DFT Simulated Raman Spectrum

The theoretical Raman spectrum of N-acetylmuramic acid was generated using density functional theory (DFT) calculations performed in Gaussian software employing the B3LYP functional with the 6-31G(d) basis set. Following geometry optimization, vibrational frequency calculations were carried out to obtain the Raman-active vibrational modes and corresponding theoretical Raman activities of the molecule.

The optimized molecular structure of N-acetylmuramic acid exhibited stable vibrational behavior without the presence of imaginary frequencies, confirming that the obtained geometry corresponded to an energetically stable configuration. The simulated Raman spectrum revealed several characteristic vibrational bands associated with the carbohydrate backbone, glycosidic linkages, acetyl functional groups, and CH-related molecular vibrations present in the NAM structure.

The calculated Raman-active modes in the lower and intermediate wavenumber regions were primarily associated with skeletal vibrations, carbohydrate ring deformations, and C–O stretching modes. Vibrational bands observed in the region between 1000–1200 cm^{-1} were attributed mainly to carbohydrate-associated stretching vibrations and ring-related molecular motions. Higher wavenumber regions showed contributions from CH bending vibrations and acetyl-associated vibrational modes. The simulated spectrum also provided detailed insight into the molecular vibrational behavior that could not be directly interpreted from the experimental spectrum alone.

To improve agreement between the theoretical and experimental Raman spectra, appropriate frequency scaling was applied during spectral comparison. The scaled theoretical frequencies demonstrated good correspondence with the experimentally observed Raman bands, supporting the validity of the computational model and the reliability of the vibrational assignments. The DFT-generated Raman spectrum therefore served as an important tool for interpreting the experimentally observed spectral features and assigning characteristic molecular vibrations of N-acetylmuramic acid.

The theoretical analysis further contributes to the development of reference vibrational spectral data for bacterial cell-wall-related biomolecules and supports future Raman and SERS-based biomarker identification studies involving peptidoglycan-associated molecular systems.

C. Experimental vs Theoretical Comparison

A comparative analysis between the experimental Raman spectrum and the DFT-simulated Raman spectrum of N-acetylmuramic acid was carried out to validate the theoretical vibrational assignments and evaluate the agreement between the observed and calculated spectral features. The experimentally obtained Raman peaks showed good correspondence with the theoretically predicted vibrational frequencies generated using Gaussian software, indicating the reliability of the computational model employed in the present study.

The comparison revealed that several experimentally observed Raman bands closely matched the scaled theoretical frequencies, particularly in the fingerprint region between 800–1500 cm^{-1} , where characteristic carbohydrate and amino sugar vibrational modes are dominant. Minor variations between experimental and theoretical peak positions were observed, which may be attributed to intermolecular interactions, experimental acquisition conditions, anharmonic effects, and limitations associated with the theoretical approximation used in DFT calculations.

TABLE I
COMPARISON OF EXPERIMENTAL AND THEORETICAL RAMAN
FREQUENCIES

Experimental Raman Shift (cm^{-1})	Theoretical Raman Shift (cm^{-1})	Proposed Vibrational Assignment
873	878	C–O stretching vibration
1092	1095	Carbohydrate ring vibration
1450	1445	CH bending vibration
1652	1648	Acetyl-associated vibration

Theoretical calculations provided valuable support for assigning experimentally observed Raman peaks to specific molecular vibrations. Characteristic bands corresponding to carbohydrate ring vibrations, C–O stretching modes, CH bending vibrations, and acetyl-associated vibrational modes showed consistent agreement between the experimental and simulated spectra. The DFT analysis also assisted in identifying overlapping vibrational contributions that were difficult to interpret solely from the experimental spectrum.

A representative comparison of selected experimental and theoretical Raman frequencies along with their proposed vibrational assignments is presented in Table I. The close agreement between the two datasets demonstrates the effectiveness of combining Raman spectroscopy with density functional theory for the vibrational characterization of biologically relevant molecules such as N-acetylmuramic acid.

The observed experimental–theoretical agreement supports the validity of the proposed vibrational assignments and establishes reliable reference Raman spectral information for isolated N-acetylmuramic acid. The combined experimental and computational approach presented in this work may further support future Raman-based bacterial biomarker studies, computational vibrational spectroscopy investigations, and SERS-assisted biosensing applications involving peptidoglycan-related biomolecules.

D. Vibrational Assignments

Detailed vibrational assignments of the experimentally observed Raman bands of N-acetylmuramic acid were performed with the assistance of the DFT-simulated Raman spectrum obtained using Gaussian software. The assignments were based on the comparison between experimentally observed Raman shifts, theoretically calculated frequencies, Raman activities, and previously reported vibrational studies of amino sugars and carbohydrate-related biomolecules available in the literature.

The Raman bands observed in the lower wavenumber region were primarily associated with skeletal molecular vibrations and ring deformation modes of the carbohydrate backbone. Vibrational bands in the intermediate spectral region between 800–1200 cm^{-1} were mainly attributed to C–O stretching vibrations, glycosidic bond-related molecular motions, and carbohydrate ring vibrations characteristic of amino sugar structures. These vibrational modes are considered important spectral signatures associated with peptidoglycan-related molecular systems.

The spectral region between 1300–1500 cm^{-1} showed prominent contributions from CH bending and deforma-

TABLE II
MAJOR RAMAN BANDS AND PROPOSED VIBRATIONAL ASSIGNMENTS

Raman Shift Region (cm^{-1})	Proposed Vibrational Assignment
800–950	Glycosidic linkage and C–O stretching vibrations
1000–1200	Carbohydrate ring vibrations and C–O–C stretching
1300–1500	CH bending and deformation vibrations
1600–1700	Acetyl-associated and coupled molecular vibrations

tion vibrations originating from the carbohydrate and acetyl functional groups present in the molecular structure of N-acetylmuramic acid. Higher wavenumber vibrational bands were associated with acetyl-related molecular vibrations and coupled vibrational modes involving oxygen-containing functional groups.

The DFT calculations significantly improved the interpretation of overlapping Raman bands by enabling visualization of the molecular vibrational behavior corresponding to individual Raman-active modes. The theoretical analysis also assisted in distinguishing closely spaced vibrational contributions that could not be clearly resolved from the experimental spectrum alone. The overall agreement between the experimental and theoretical spectral features supports the reliability of the proposed vibrational assignments.

A summary of the major Raman bands and their corresponding vibrational assignments is presented in Table II.

The vibrational assignments obtained in the present study contribute to the establishment of reference Raman spectral fingerprints for isolated N-acetylmuramic acid. These findings may support future Raman spectroscopy studies involving bacterial cell-wall components, peptidoglycan-related biomolecules, and Raman/SERS-based bacterial biomarker detection systems.

E. Biomarker Relevance

N-acetylmuramic acid is an essential structural component of bacterial peptidoglycan and plays a critical role in maintaining the integrity and mechanical stability of bacterial cell walls. Since peptidoglycan is a unique feature of bacterial systems and is absent in mammalian cells, molecules such as N-acetylmuramic acid have gained significant importance in bacterial biomarker research and infection-related diagnostic studies. The identification and characterization of NAM-specific vibrational signatures may therefore provide valuable information for future bacterial detection and biosensing applications.

The experimentally obtained Raman fingerprints and corresponding DFT-assisted vibrational assignments presented in this work contribute toward the development of reference spectral data for bacterial cell-wall-associated biomolecules. The identified Raman-active vibrational modes associated with carbohydrate ring structures, glycosidic linkages, C–O stretching vibrations, and acetyl functional groups may serve as important spectral markers for peptidoglycan-related molecular identification.

Several Raman and SERS-based bacterial sensing studies have previously focused on whole-cell bacterial analysis, where spectral contributions from multiple biomolecules

often overlap and complicate interpretation. In contrast, the present work focuses specifically on isolated N-acetylmuramic acid, enabling a more detailed understanding of its individual molecular vibrational characteristics. Such molecule-specific spectral characterization is important for improving the interpretation of complex bacterial Raman spectra and for supporting future computational and experimental bacterial classification studies.

The combination of experimental Raman spectroscopy and density functional theory analysis further enhances the reliability of biomolecular vibrational assignments and provides molecular-level insight into Raman-active modes associated with bacterial cell-wall components. The reference spectral information obtained from the present study may therefore support future investigations involving Raman- and SERS-based bacterial diagnostics, biomarker identification, peptidoglycan analysis, and machine learning-assisted spectral classification of bacterial systems.

In addition, the present work may contribute to future biosensor development studies involving bacterial infection monitoring, rapid microbial identification, and sepsis-related biomarker detection. The experimentally validated Raman spectral fingerprints of isolated N-acetylmuramic acid generated in this study may serve as foundational vibrational reference data for advanced analytical and biosensing applications involving bacterial cell-wall-associated biomolecules.

IV. CONCLUSION

In the present study, the vibrational characteristics of pure N-acetylmuramic acid powder were investigated using experimental Raman spectroscopy combined with density functional theory (DFT)-based computational analysis. Experimental Raman spectra were successfully acquired under optimized measurement conditions, and characteristic vibrational bands associated with carbohydrate ring vibrations, C–O stretching modes, CH bending vibrations, and acetyl-related molecular vibrations were identified and analyzed.

The experimentally obtained Raman spectra showed good agreement with the theoretically simulated Raman frequencies generated using Gaussian software employing the B3LYP/6-31G(d) computational approach. The combined experimental and theoretical analysis enabled detailed vibrational assignments of the major Raman-active modes and provided molecular-level insight into the vibrational behavior of N-acetylmuramic acid. The close correspondence between the experimental and simulated spectra supports the reliability of the computational model and validates the proposed spectral assignments.

The present work establishes reference Raman spectral fingerprints of isolated N-acetylmuramic acid and contributes experimentally validated vibrational information relevant to bacterial cell-wall-associated biomolecules. The findings of this study may support future investigations involving Raman and SERS-based bacterial biomarker detection, peptidoglycan analysis, computational vibrational spectroscopy, and biosens-

ing applications related to bacterial diagnostics and infection monitoring.

Overall, the study demonstrates the effectiveness of combining Raman spectroscopy with DFT simulations for the characterization of biologically important amino sugar molecules and provides a foundational framework for future experimental and computational investigations involving bacterial biomarker systems.

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